

Lean Canvas

1. Full disclosure statement of prior participation of the Conrad Challenge including past team members names who may not still be with your team. If your innovation was used previously in another similar competition share those details similarly. Details of similar innovation challenges you plan to submit the same innovation to this school year.

Two of our teammates, Siddarth Mohan and Sahasra Beerala, have participated in the Conrad Challenge in 2023-2024. However, the innovation used that year is completely different from our current idea. Additionally, we plan to submit our current innovation to the Diamond Challenge.

1. High-Level Concept: What is the tagline of your innovation, in 10 words or less? (10 words)- siri

Deorbit origami: more drag, less debris.

2. Problem: What is the customer need your innovation will address? Is there a social or environmental challenge you aim to take on? (40 words)- armani

More than 9,000 metric tons of orbital debris are currently in Earth's orbit. At the current rate, the number is expected to significantly increase. However, many current solutions are expensive, delicate, and aerodynamically unstable.

3. Existing Alternatives: How is this problem solved today? Consider other products in the market. (40 words)- siri

Today, there are current drag-based designs, like the Spinnaker Series and ADEO Series, in the market. However, the thin materials used to create these products are delicate and may not survive long missions.

4. Solution: What are the key characteristics of your innovation? (40 words)- sid

Our solution reduces space debris by using a mechanism that increases aerodynamic drag on LEO objects, causing passive deorbit. Inspired by origami, the design features a tessellation-based structure that unfolds from a compact rocket volume to clear future debris.

5. Unique Value Proposition: What makes your innovation different from what's already in the market? (40 words)- sid

Current drag-based solutions use thin material that may fail in the harshness of space. Inspired by origami, the design features a tessellation-based structure that unfolds from a compact rocket volume and is much more robust.

6. Sustainable Advantage: Why will it be difficult for others to copy you? (40 words)- abhi

The usage of origami deployable structure design created out of aluminized PET (Mylar), Dyneema, and the coilable CFRP tape-spring booms is a more robust and durable solution. It integrates specific folding and provides reliable deployments.

7. Customer Segments: Who is the target audience? Who is served by your innovation? Describe them. (40 words)- siri

We plan to target satellite operators, including small satellite and CubeSat manufacturing, as well as space agencies/regulators. These target audiences seek efficient, inexpensive, and compliant solutions for satellite deployment and disposal.

8. Early Adopters: Who will be your very first customers? Describe them. (40 words)- sid

Our initial customers will be small satellite manufacturers like Planet Labs, as they operate in LEO and must comply with the 25-year deorbit rule. Given the small size of their satellites, they need simple solutions, making them ideal early adopters.

9. Channels: How will you sell your innovation to your customers? How will you deliver it? (40 words)- armani

Our deorbit system will be sold as a ready-to-install component to satellite manufacturers. Distribution will occur through online sales, partnerships, and industry expos. This allows us to integrate our product with the needs of the consumer to accomplish their missions.

10. Key Metrics: What are the most important numbers that track your success? - abhi

Some of the most important numbers that will track our success are the number of deorbit origami sold, the amount of government investment, and the number of companies/organizations investing.

11. Cost Structure: What are your most significant costs? (40 words)- abhi

Our recurring variable costs would be the price of the aerospace grade materials. Our materials include aluminized PET (Mylar) and Dyneema, which form the deployable surface, and the coilable CFRP tape-spring booms that creates the structural support.

12. Revenue Streams: How will you make money to fund your operations on an ongoing basis? (40 words)- armani

Our revenue is generated through direct sales to satellite operators, licensing and partnerships producing royalties, government and public grants/contracts, private investments, service contracts with satellite operators, and sustainability certification partnerships that create recurring fees, ensuring long-term revenue.